Abstract

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Inventory Optimization Under Uncertainty: Adaptive Decision-Making with Reinforcement Learning in the Quality-Dependent Newsvendor Problem.
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The present thesis investigates the application of reinforcement learning (RL) to stochastic inventory management, focusing on the Quality-Dependent Newsvendor Problem (QDNP). Moreover, a data-driven variant incorporating kernel density estimation (KDE) is considered. The QDNP models inventory decisions under uncertain demand and product quality deterioration, integrating donation as a strategic decision. Donations reduce waste by redirecting surplus products to food banks and other charitable organizations, supporting social and environmental responsibility. Also, they improve profits by leveraging corporate social responsibility (CSR) incentives, which are appealing to socially and ecologically conscious consumers, and avoid disposal costs.

To this end, a state-of-the-art deep reinforcement learning (DRL) approach, namely the Soft Actor-Critic (SAC) algorithm, is employed to maximize expected profit by simultaneously optimizing decisions on order quantities, pricing, and donations across multiple real-world store instances. In this context, SAC is introduced as an alternative to classical stochastic optimization methods, with the additional potential for knowledge transfer. The results obtained indicate that SAC can closely approximate the analytical solution, while maintaining practical accuracy. A KDE-based extension is also studied, which empirically approximates the demand component distribution instead of assuming full prior knowledge. The corresponding results showed slightly increased approximation errors but remained within acceptable thresholds. Further-

more, zero-shot policy transfer showed that pretrained SAC policies can adapt to moderate changes in demand and cost parameters, although retraining is required for larger deviations.

Using SAC within a controlled inventory problem, this study demonstrates how sequential decision-making, and stochastic uncertainty can be addressed through RL. The findings provide a concrete example of RL applied to inventory management and contribute to the growing intersection of reinforcement learning and operations management, assessing its potential for integrating data-driven components and dynamic decision-making methods.

Keywords: Quality-Dependent Newsvendor Problem, Deep Reinforcement Learning, Soft Actor-Critic, Stochastic Optimization, Policy Transfer, Data-Driven Decision Making, Inventory Management, Kernel Density Estimation